Errata

On Coppock and Green (2016), *American Journal of Political Science* 60(4)

This note lists ten corrections to "Is Voting Habit Forming? New Evidence from Experiments and Regression Discontinuities," published in the *American Journal of Political Science* 60(4):1044–1062, doi 10.1111/ajps.12210. **None of them changes any conclusion of the paper.** One is a standard error, one is a row of an appendix table, six are sentences or a heading that miscount or misdescribe something the article prints correctly, and the last is an entry of the reference list. Everything else in the 1,329 published table cells this note's supporting analysis reads off the page reproduces from the deposited replication archive (10.7910/DVN/ALZVAW).

Every number in the corrected sentences and in the notes beneath them is computed from the deposited archive at the time this document is rendered. The code and output supporting the corrections are at github.com/active-maintenance-aec/coppock_green_2016.

1. The standard error of the primary-on-primary estimate

> The average estimated primary-on-general CACE is 0.0086 (SE = 0.0236), while the average estimated primary-on-primary CACE is 0.0068 (SE=**0.0010**).

Corrected:

> The average estimated primary-on-general CACE is 0.0086 (SE = 0.0236), while the average estimated primary-on-primary CACE is 0.0068 (SE=**0.0099**).

The estimate and the other three figures in the sentence reproduce exactly. As printed, a standard error of 0.0010 would put the primary-on-primary estimate 6.8 standard errors from zero, which contradicts the sentence's own point that inducement to vote in a primary has negligible downstream effects. Recomputed, the estimate is 0.69 standard errors from zero, and is indistinguishable from zero as the passage says.

2. The count of midterm-on-presidential estimates

> The pattern of midterm-on-presidential estimates argues strongly for habit formation as well. Here, **50** out of **54** estimates are positive.

Corrected:

> The pattern of midterm-on-presidential estimates argues strongly for habit formation as well. Here, **51** out of **55** estimates are positive.

The lower panel of Table 4 prints 55 state estimates, not 54; the upper panel prints 54, which is the figure the preceding sentence correctly gives for the presidential-on-midterm effects. The two panels differ because Oregon enters the midterm-on-presidential series in 2006–08 and has no 2004–06 counterpart in the presidential-on-midterm series. Of the 55 estimates, four are negative, so 51 are positive. Table 4 itself is correct.

3. Battleground status among the strongest presidential-on-presidential estimates

> The top panel of Table 5, for example, reports 15 estimates of the presidential-on-presidential downstream effect for 2008–12. The top **four** of the five strongest estimates (Arkansas, Nevada, Connecticut, Missouri, and New Jersey) are found in nonbattleground states.

Corrected:

> The top panel of Table 5, for example, reports 15 estimates of the presidential-on-presidential downstream effect for 2008–12. The top **three** of the five strongest estimates (Arkansas, Nevada, Connecticut, Missouri, and New Jersey) are found in nonbattleground states.

Under the article's own coding, a state is a presidential battleground when the two-party vote share difference in the downstream election is ten points or fewer. Nevada at 6.8 points and Missouri at 9.6 points clear that threshold in 2012, so three of the five strongest estimates are in nonbattleground states rather than four. Ten of the fifteen states with a 2008-12 estimate are nonbattleground, so 3.33 of the five strongest are expected to be nonbattleground by chance alone; three is the modal outcome, at $p = 0.40$, and the published four or more occurs 43 percent of the time. The corrected count is what chance gives, which is what the passage asserts two sentences earlier: "In fact, we see little difference."

4. The change in the battleground coefficient between columns 2 and 3 of Table 7

> When we include the close midterm elections in the definition of battleground, this pattern does not persist—the coefficient drops by a factor of **three** and is no longer significant.

Corrected:

> When we include the close midterm elections in the definition of battleground, this pattern does not persist—the coefficient drops by a factor of **two** and is no longer significant.

Table 7 prints 0.0322 in column 2 and 0.0155 in column 3, a ratio of 2.1. Both cells reproduce exactly; only the description of the ratio is wrong, and the coefficient is indeed no longer significant.

5. The number of states in Table A2

> Table A2 presents the estimated number of net migrants, the estimated number of compliers in 2008 (computed using a 365-day window), and the associated bias according to Equation A10 for **sixteen** states.

Corrected:

> Table A2 presents the estimated number of net migrants, the estimated number of compliers in 2008 (computed using a 365-day window), and the associated bias according to Equation A10 for **fifteen** states.

The table has fifteen rows, one for every state with a 2008-on-2012 estimate. Colorado and Michigan have no such estimate, so the analysis covers fifteen of the seventeen states.

6. Appendix Table A2's Iowa row

Table A2's CACE 08-12 column holds each state's estimated effect of voting in 2008 on voting in 2012, and for fourteen of its fifteen states it agrees with Tables 5 and 6 exactly. For Iowa it does not: the appendix prints 0.086, where Tables 5 and 6 both print 0.081. The bias estimate and the corrected CACE in that row follow from the wrong figure, so all three cells move. The corrected table:

Table 1: Table A2 corrected: Migrants, Number of Compliers, and Associated Bias

State	Net Migrants	N_c 2008	CACE 08-12	Bias Estimate	Corrected CACE
PA	-1033	95175	0.121	-0.001	0.122

State	Net Migrants	N_c 2008	CACE 08-12	Bias Estimate	Corrected CACE
IA	-517	5318	0.081	-0.007	0.088
CT	-183	22413	0.161	-0.001	0.162
NV	-182	12465	0.174	-0.003	0.177
RI	-117	6867	0.113	-0.002	0.115
OK	-50	15584	0.138	-0.000	0.138
NJ	8	51983	0.155	0.000	0.155
MO	17	35535	0.155	0.000	0.155
MT	50	6770	0.111	0.001	0.110
KY	133	21964	0.075	0.000	0.075
OR	134	17853	0.108	0.001	0.107
AR	183	11778	0.200	0.003	0.197
NY	292	79908	0.068	0.000	0.068
FL	641	88519	0.105	0.001	0.104
IL	750	58991	0.080	0.001	0.079

The changed artifact is the Iowa row alone; every other cell is as published. The deposited code for this table types its fifteen CACEs in as constants rather than reading them from the regression discontinuity output the rest of the paper uses, and the constant for Iowa is not the estimate that analysis produces.

7. The cross-reference for the general-on-general election pairs

> Across the 384 general-on-general election pairs (the pairs reported in **Table 6** of the main text), the average estimated CACE is 0.101 with a standard error of 0.007.

Corrected:

> Across the 384 general-on-general election pairs (the pairs reported in **Table 7** of the main text), the average estimated CACE is 0.101 with a standard error of 0.007.

Table 6 reports 86 pairs, every upstream general election on downstream voting in 2012. The 384 general-on-general pairs are the set Table 7 pools, and Table 7's own N row prints that figure in all five columns. Both averages in the sentence reproduce exactly.

8. The first panel of appendix Table A4

> **Arizona 2013**

Corrected:

Arkansas 2013

Tables A4 and A5 give descriptive statistics for one panel per voter file, nineteen in all: the seventeen states the article analyses plus the two historical files for Florida and Missouri. Every other panel corresponds to one of those states. Arkansas is one of the seventeen, appears in Tables 4, 5, 6 and A2, and has no panel of its own; Arizona appears in no table in the article and is nowhere in the deposited data, whose states are AR, CO, CT, FL, IA, IL, KY, MI, MO, MT, NJ, NV, NY, OK, OR, PA, and RI. The changed artifact is the panel heading alone; every figure beneath it is as published.

9. The leverage adjustment behind the discontinuity standard errors

Table 4, Iowa, 2004-06 (Presidential on Midterm), SE: **0.049**

Table 5, Kentucky, 2008-12 (Presidential on Presidential), SE: **0.022**

Table 6, Kentucky, 2008-12 (lower), SE: **0.022**

Table 6, Meta-Analysis, 2002-12 (lower), estimate: **0.271**

Table 7, Presidential upstream, (5), SE: **0.0207**

Table A6, Second-order Polynomial, 90 Days (No additional controls), estimate: **-0.028**

Table A6, Third-order Polynomial, 635 Days (Controls for lagged vote totals), SE: **0.009**

Table A7, Second-order Polynomial, 90 Days (Controls for lagged vote totals), estimate: **-0.006**

Table A7, Third-order Polynomial, 545 Days (No additional controls), estimate: **0.101**

Table A7, Third-order Polynomial, 90 Days (Controls for lagged vote totals), estimate: **0.002**

Table A7, Third-order Polynomial, 90 Days (No additional controls), estimate: **-0.019**

Corrected:

Table 4, Iowa, 2004-06 (Presidential on Midterm), SE: **0.048**

Table 5, Kentucky, 2008-12 (Presidential on Presidential), SE: **0.021**

Table 6, Kentucky, 2008-12 (lower), SE: **0.021**

Table 6, Meta-Analysis, 2002-12 (lower), estimate: **0.272**

Table 7, Presidential upstream, (5), SE: **0.0206**

Table A6, Second-order Polynomial, 90 Days (No additional controls), estimate: **-0.027**

Table A6, Third-order Polynomial, 635 Days (Controls for lagged vote totals), SE: **0.008**

Table A7, Second-order Polynomial, 90 Days (Controls for lagged vote totals), estimate: **-0.007**

Table A7, Third-order Polynomial, 545 Days (No additional controls), estimate: **0.100**

Table A7, Third-order Polynomial, 90 Days (Controls for lagged vote totals), estimate: **0.001**

Table A7, Third-order Polynomial, 90 Days (No additional controls), estimate: **-0.020**

Every standard error in the discontinuity tables is heteroskedasticity-robust, and the deposited code computes it as coeftest(fit, vcov = vcovHC) on a two-stage least squares fit from the AER package. The sandwich package carries no leverage convention of its own: it calls hatvalues() on the fit it is handed. AER's method for a 2SLS fit returns the diagonal of the asymmetric matrix that maps the outcome onto its fitted values, and Belsley, Kuh, and Welsch (1980) considered exactly that quantity for 2SLS diagnostics and recommended against it, on the ground that the diagonal of an asymmetric matrix is not a leverage. The ivreg package, which now carries the implementation AER once did, follows that recommendation and uses the second-stage projection, whose diagonal lies between zero and one by construction. The maintained pipeline follows the later convention. On the Iowa 2004-06 pair the two give 0.048544 and 0.048498, either side of the boundary at which the third decimal turns. The difference runs to about a part in a thousand, so the published figure stands at the precision these tables print in 926 of 937 cells. The eleven that move are of two kinds. Three are a single state's own standard error, each sitting on a rounding boundary. The other eight are pooled quantities that inherit those standard errors through a weight: Table 6's meta-analytic estimate is precision-weighted by them, Table 7's models are weighted by their inverse squares, and every cell of appendix Tables A6 and A7 is a fixed-effects meta-analysis across the same states, whose weights are the same standard errors. That is how the convention reaches a point estimate, which each state's own estimate is immune to. No sentence in the article or the appendix quotes any of the eleven. The appendix's six sit in the tables its Section 2 offers as evidence that the estimates are quite robust to specification, and the corrected values leave that reading exactly where it was: the 2008-on-2012 estimates still cluster between 0.10 and 0.12, the 2006-on-2010 estimates still fall mostly between 0.07 and 0.15, and the standard errors still shrink as the window widens.

The reference list

The entry below corrects the article's reference list. It was found by an audit that sends every printed entry whole to Crossref and checks the authoritative record back into it, and it is verified against the record named in the note. No cited work is missing and no citation in the text points at the wrong work.

10. Atkinson and Fowler (2014) prints the last page of its range without the first

Published, p. 1061:

> Atkinson, Matthew D., and Anthony Fowler. 2014. "Social Capital and Voter Turnout: Evidence from Saint's Day Fiestas in Mexico." British Journal of Political Science **44(1): 59**.

Corrected:

> Atkinson, Matthew D., and Anthony Fowler. 2014. "Social Capital and Voter Turnout: Evidence from Saint's Day Fiestas in Mexico." British Journal of Political Science 44(1): 41–59.

Crossref, doi 10.1017/s0007123412000713: pages 41–59. The reference list abbreviates the second number of a range elsewhere, so a reader has no way to read "59" as anything but a single page.